

Course	Mandatory Activity	Skills Developed
Problem Solving using C	30 practice problems + industry-specific final test; Codethon; Mini Project	C programming, problem solving, algorithmic thinking, debugging, coding speed, competitive programming, application development
Web Essentials	Personal portfolio; Responsive web design mini project	HTML/CSS, responsive design, web development, UI implementation, portfolio development
Maker Space	3D printing design; Electronics project	3D modelling, prototyping, electronics, hardware problem solving, design thinking
Introduction to Artificial Intelligence	Coursera: ML with Scikit-learn, PyTorch & Hugging Face; GATE practice	AI/ML fundamentals, Scikit-learn, PyTorch, Hugging Face, model implementation, analytical thinking
Data Structures using C	Practice problems, industry final test, Codethon, GATE practice	Data structures, C programming, algorithmic problem solving, coding

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		efficiency, competitive programming
Digital Principles & Computer Organization	Quiz; Digital Logic Design Challenge	Digital logic, logical reasoning, circuit design, computer organization fundamentals
Foundations of Data Science	Practice problems, final test, Codethon, mini project	Python/data-science problem solving, data analysis, analytical thinking, project development
ReEngineering for Innovation	Hardware project design + implementation	Product design, prototyping, hardware implementation, innovation, project management